

THE UNIVERSITY OF AUCKLAND

SEMESTER TWO, 2023

Campus: City

COMPUTER SCIENCE

(Time allowed: TWO hours)

NOTE:

Write your name here:

Write your UPI here:

Write your ID# here:

- This is a closed-book assessment.
- There are 30 questions. The total number of marks is 100.
- This exam is worth 40% of your final grade.

1 Locality-Sensitive Hashing

[25 marks]

Given four documents S_1, S_2, S_3, S_4 and a customized query document q :

$$S_1 = \{c, f\}, S_2 = \{a, b\}, S_3 = \{d, e\}, S_4 = \{a, c, e\}, q = \{a, c, e\};$$

1.1 Computing MinHash Signatures

[10 marks]

- (a) Generate the bit-vector representation for $\{S_1, S_2, S_3, S_4, q\}$ in shingle space $\{a, b, c, d, e, f\}$. [3 marks]
- (b) Generate the MinHash matrix for $\{S_1, S_2, S_3, q\}$ using the following four MinHash functions. [5 marks]

$$h_1(x) = (2x + 1) \mod 6$$

$$h_2(x) = (3x + 2) \mod 6$$

$$h_3(x) = (5x + 2) \mod 6$$

- (c) Among the hash functions, h_1, h_2, h_3 , which one gives the true simulated permutation? [1 marks]
- (d) Consider the query q and estimate the signature-based Jaccard similarities: $J(q, S_1)$. [1 marks]

1.2 Tuning Parameters for rNNS

[10 marks]

Recall the collision probability (i.e., S-curve) given the number of bands b and the number of rows per band r as follows:

$$Pr(s) = 1 - (1 - s^r)^b$$

Consider three sets of parameters $(r=3, b=10)$, $(r=6, b=20)$, $(r=5, b=50)$. The collision probabilities for similarity s in range of $[0, 1]$ for each (r, b) are provided accordingly in the table below. Please answer the following two questions.

- (a) Which settings give at most 5% of false negatives for any 70%-similar pairs? Please briefly explain your answer. [5 marks]
- (b) Which settings give at most 15% of false positives for any 30%-similar pairs? Please briefly explain your answer. [5 marks]

s	($r = 3, b = 10$)	($r = 6, b = 20$)	($r = 5, b = 50$)
0	0	0	0
0.1	0.00996	1.9×10^{-5}	4.9×10^{-4}
0.2	0.07718	0.00128	0.01588
0.3	0.23945	0.01448	0.11454
0.4	0.48387	0.07880	0.40229
0.5	0.73692	0.27019	0.79555
0.6	0.91227	0.64541	0.98253
0.7	0.98502	0.91819	0.99999
0.8	0.99923	0.99771	1
0.9	0.99999	0.99999	1
1	1	1	1

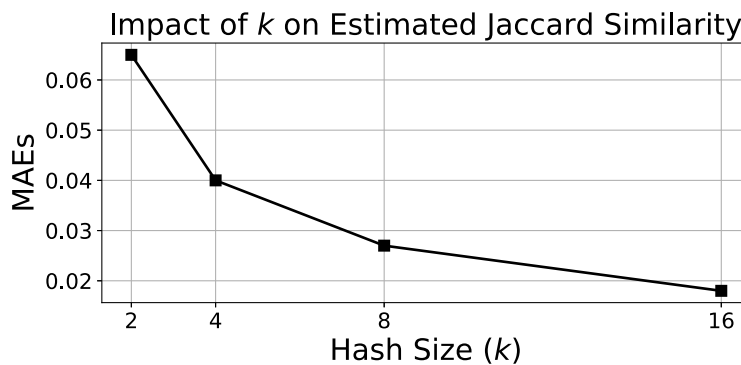
1.3 c -Approximate Randomized rNNS

[5 marks]

Recall that a family of functions H is called (d_1, d_2, p_1, p_2) -sensitive with collision probability $p_1 > p_2$ and $c > 1$ if the following conditions hold for any uniformly chosen $h \in H$ and $\forall x, y \in U$:

- If $d(x, y) \leq r$, $\Pr[h(x) = h(y)] \geq p_1$ for similar points, and
- If $d(x, y) \geq cr$, $\Pr[h(x) = h(y)] \leq p_2$ for dissimilar points.

- (a) How can we further transform the upper bound probability approaching 1? How can we further transform the lower bound probability approaching 0? [2 marks]
- (b) Consider a set of movie plot articles (D), and a set of queries (Q). You have constructed Locality-Sensitive Hashing (LSH) frameworks for varying hash sizes with $k \in \{2, 4, 8, 16\}$. To evaluate the impact of k on Jaccard similarity estimation, you computed the mean absolute error (MAE) for pairs of articles (d_i, q) , where $d_i \in D$ and $q \in Q$, and obtained the MAE plot below. Please comment on your MAE plot. (Note: You can assume that your implementation is correct.) [3 marks]



2 Data Stream Algorithms

[25 marks]

2.1 Bloom Filter

[6 marks]

- (a) Consider the following statement: “*The Bloom filter can determine the existence of items, allowing the possibility of false positive responses and false negative responses.*” Is it true, false or unknown? Please briefly explain your answer. [3 marks]
- (b) Name two approaches to reduce the possibility of false positives. Please briefly explain your answer. [3 marks]

2.2 Misra-Gries Algorithm

[5 marks]

- (a) Consider the given data stream. Implement the Misra-Gries algorithm with $k = 4$ counters. Provide a comprehensive summary, including the elements and their corresponding counter values, once the execution of the algorithm is completed. [3 marks]

$$S = \{45, 13, 14, 45, 7, 10, 45, 6\}$$

- (b) What would the theoretical number of decrement steps be triggered after processing the above stream in question (a)? [2 marks]

2.3 Count Sketch Algorithm

[14 marks]

Consider the same data stream S and hash functions below. Using the given sign hash functions below, apply the Count Sketch algorithm and present the following:

- (a) Hash Table. [5 marks]

$$s_i(x) = \begin{cases} 0, & -1 \\ 1, & +1 \end{cases}$$

Elements	45	13	14	7	10	6
$h_1(x) = x \bmod 3$						
$h_2(x) = (2x + 1) \bmod 3$						
$h_3(x) = (3x + 1) \bmod 3$						
$s_1(x) = ((2x + 1) \bmod 3) \bmod 2$						
$s_2(x) = ((3x + 1) \bmod 3) \bmod 2$						
$s_3(x) = ((x + 2) \bmod 3) \bmod 2$						

- (b) Counter Matrix. [5 marks]
- (c) Final estimated frequency of each element once the execution of the algorithm is completed. Please comment on your estimated frequency against the true frequency. [4 marks]

3 Algorithms for Graphs [25 marks]

3.1 Web Graphs [12 marks]

Given the following raw adjacency matrix $\mathbf{A}$:

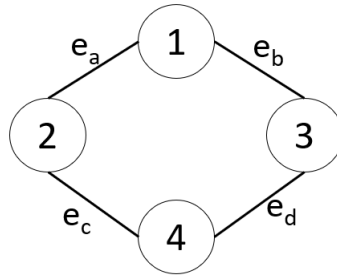
$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Convert $\mathbf{A}$ into its corresponding column-stochastic matrix $\mathbf{M}$. [4 marks]
- (b) In the lecture, we have shown that the PageRank $\mathbf{r} = \mathbf{M} \cdot \mathbf{r}$ is the eigenvector of the column-stochastic matrix $\mathbf{M}$ corresponding to eigenvalue $\lambda = 1$. Compute the PageRank of all nodes in the above graph using the eigen-equation and report the values of the $\mathbf{r}$ vector. [4 marks]
- (c) The power iteration algorithm is an iterative method to solve PageRank. Explain: 1) how does the algorithm perform updates in each iteration? 2) what are the stop criteria and required initializations? [4 marks]

3.2 Community Detection [8 marks]

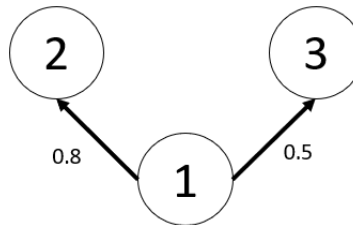
Given the following undirected graph:

- (a) Compute the edge betweenness of each edge ($\{e_a, e_b, e_c, e_d\}$) using Brandes' algorithm, only for the case when node 1 is taken as the root node. [4 marks]
- (b) Apart from edge betweenness and modularity maximization, spectral graph partitioning is another method for community detection that works by solving the problem $\min_{\mathbf{y} \in \mathbb{R}^n} \mathbf{y}^T \mathbf{L} \mathbf{y}$, with $\mathbf{L}$ the Laplacian matrix. Explain: 1) how is the Laplacian matrix calculated, 2) how can eigenvalue decomposition be used to solve this minimization problem. [4 marks]



3.3 Influence Maximization [5 marks]

Given the following directed graph:



- Consider the set $S = \{1\}$ as the seed set. Compute the contingency table for the graph. [3 marks]
- Using the contingency table calculated in the previous step, compute the influence spread under the Independent Cascade (IC) model. [2 marks]

4 Recommender Systems [25 marks]

4.1 Memory-Based Collaborative Filtering [10 marks]

Given the following user-interaction matrix of users $\{u_1, u_2, u_3, u_4, u_5\}$ and items $\{p_1, p_2, p_3, p_4\}$:

- Based on these ratings from users to items, which user is most similar to user u_2 if you consider cosine similarity? How does this information help you to provide recommendations to user u_2 if you were to use the user-based collaborative filtering algorithm? [4 marks]
- Calculate: 1) the global bias (bg), 2) the user bias for user u_2 ($b_{u_2}^{(user)}$), and 3) the item bias for item p_1 ($b_{p_1}^{(item)}$). [3 marks]

	p_1	p_2	p_3	p_4
u_1	5	?	2	?
u_2	3	4	?	?
u_3	4	?	1	4
u_4	1	3	?	?
u_5	5	4	?	?

- (c) Suppose that you used one of the memory-based recommendation algorithms learned in lectures to calculate predicted ratings of users over item p_1 . You obtained the following comparison of predicted vs. actual ratings:

User	Item	Groundtruth/ actual rating r	Predicted rating $\hat{r}$
u_1	p_1	5	4
u_2	p_1	3	2
u_3	p_1	4	5
u_4	p_1	1	4
u_5	p_1	5	3

Calculate the root mean squared error (RMSE) metric of the above table (report the RMSE using two decimal digits). [3 marks]

4.2 Model-based Collaborative Filtering [8 marks]

In the lectures, we discussed about the Latent Factor Model (LFM) as a model-based technique for approximating the user-interaction matrix, and therefore learning how to make predictions on unknown ratings. Considering the regularized optimization problem of LFM (without bias):

$$\min_{\mathbf{P}, \mathbf{Q}} \sum_{r_{ij} \in R} (r_{ij} - \mathbf{q}_i^T \cdot \mathbf{p}_j)^2 + \lambda_1 \sum_{i \in U} \|\mathbf{q}_i\|_2^2 + \lambda_2 \sum_{j \in P} \|\mathbf{p}_j\|_2^2$$

Explain, in a few sentences:

- (a) What is this optimization problem of about? [3 marks]
- (b) How can Stochastic Gradient Descent help solve this problem? [3 marks]

- (c) What is the advantage of a model-based technique such as LFM as opposed to memory-based techniques such as user-based or item-based collaborative filtering? [2 marks]

4.3 Context-aware Recommendations [7 marks]

Suppose you are asked to build a context-aware system to recommend events. Users $\{u_1, u_2, u_3, u_4\}$ attend events from $\{e_1, e_2, e_3\}$ in groups. Events are held in one of the two stadiums s_1 and s_2 . After attending, users assign ratings to events. The following table shows the transactions:

Transaction ID	Group of users	User	Event	Stadium	Rating
1	A	u_1	e_1	s_1	5
2	A	u_2	e_1	s_1	5
3	B	u_3	e_2	s_1	3
4	B	u_4	e_2	s_1	3
5	C	u_2	e_3	s_2	4
6	C	u_4	e_3	s_2	4
7	D	u_3	e_1	s_2	3
8	D	u_4	e_1	s_2	3

- (a) Define an encoding for groups, users, events and stadiums, and construct the input feature vectors for the factorization machine using the event transactions in the above table. [4 marks]
- (b) How can the information in these feature vectors along with the information about ratings be used for prediction of future ratings? Explain your answer. [3 marks]